`ironmc` / `rmc-ab-initio` project glossary

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Working vocabulary of the ab-initio DiagMC project. Terms are grouped into **physics** (Section 1) and **software engineering** (Section 2); overloaded words that mean different things in different contexts are disambiguated in Section 2.7. The methodology note `latex/ab-initio-diagmc/ab-initio-diagmc.t` is the authoritative physics reference; this file only defines the words.

1 Physics

1.1 Model and observables

Polaron. An electron dressed by the lattice distortion (phonon cloud) it drags along. The project computes the *electron* polaron of LiF: one carrier at the conduction-band minimum plus its phonon cloud.

Formation energy E_0. The polaron ground-state energy relative to the bare band edge, $E_0 = E_{\text{polaron}} - \varepsilon_{\text{CBM}}$. Negative; the headline observable of every production run. LiF references: -0.2479 eV on the 20^3 grid, -0.408 eV extrapolated (FEP-DMC; -0.538 eV with the remote-band correction). Ours: -0.4063 ± 0.0010 eV extrapolated at $\tau = 240$ eV^{-1}.

Fröhlich model. The textbook continuum polaron: parabolic electron band, one dispersionless LO phonon, vertex $|g(\mathbf{q})|^2 \propto \alpha/q^2$. Serves two roles here: the analytic limit LiF is close to (so an analytic cross-check exists), and the engine-validation model (see *stage*, Section 2.1).

Coupling constant α. The dimensionless Fröhlich electron–phonon coupling. $\alpha_{\text{LiF}} \approx 4.76$ (intermediate-to-strong coupling), computed from m^*, ω_{LO}, ε_∞, ε_0. Weak-coupling target: $E^{(2)} \approx -\alpha\hbar\omega_{\text{LO}}$.

LO / TO phonons, LO–TO splitting. Longitudinal / transverse optical branches. In a polar crystal the long-range dipole field stiffens the LO branch at Γ (LiF: LO 18.45 THz vs TO 8.12 THz). The splitting is carried entirely by the long-range part of the dynamical matrix, which EPW subtracts and the code must restore.

Imaginary time τ. The evolution variable of the zero-temperature imaginary-time formulation. With $\hbar = 1$ and energies in eV, τ has units eV^{-1}. Diagrams live on a τ interval $[0, \tau_{\max}]$.

Green's function $G(\tau)$. The imaginary-time electron propagator whose large-τ decay $G(\tau) \sim Ze^{-(E_0-\mu)\tau}$ encodes the ground-state energy. What the decay estimator fits; *not* the same object as the self-energy.

Chemical potential μ. A reweighting shift $e^{+\mu\tau}$ applied to the electron propagator to keep diagram weights bounded. In the decay estimator it must bracket E_0 from below (P9 of the methodology note); in the fixed-τ thermodynamic ensemble it drops out entirely.

Units (physical vs natural). The ab-initio model runs in physical units: energies eV, time eV^{-1}, momenta in fractional (crystal) coordinates on a Γ-centered grid. Fröhlich toy runs use polaron natural units $\hbar = m = \omega_{\mathrm{LO}} = 1$. Ab-initio inputs arrive in Rydberg (≈ 13.6057 eV) or THz and are converted on load.

High-symmetry points / symmetry paths (Γ, X, L, W, ...). Points of the fcc BZ left invariant (modulo reciprocal-lattice vectors) by a nontrivial subgroup of the cubic point group O_h, and the conventional 1D tours through them along which dispersions are plotted and effective masses quoted (e.g. $\Gamma \to X$ = fractional ray $(j/n, 0, j/n)$, a Cartesian cube axis). For the electron polaron the quadratic term of $E_0(\mathbf{p})$ at Γ is symmetry-forced isotropic (a scalar m^*); directions differ first at quartic order. Full background and reading list: `latex/symmetry-paths/symmetry-paths.tex`.

Effective mass m^*, CBM. Conduction-band minimum and its curvature mass. LiF's CBM is single and non-degenerate at Γ (why LiF is the first material); m^* enters α and the q-unit bridge of the reduced Fröhlich comparison.

Second-order energy $E^{(2)}$. The lowest-order (one-phonon) perturbative polaron energy, $E^{(2)}(\mathbf{k}) = -\frac{1}{N_q}\sum_{\mathbf{q}\nu} |g(\mathbf{k},\mathbf{q},\nu)|^2/(\varepsilon_{\mathbf{k}+\mathbf{q}} - \varepsilon_{\mathbf{k}} + \omega_{\mathbf{q}\nu})$. Cheap to evaluate exactly; used as the vertex-scale check and as a (biased) scanning proxy for compression error — it overestimates the in-sampler systematic by 3–10×, so it never decides a budget (see *gate*).

Lang–Firsov / dispersionless model. Exactly solvable lattice polaron ($\varepsilon(\mathbf{k}) = \text{const}$), $E_0 = -\bar{g}^2/\omega$ at all couplings. Used as an analytic validation target for the thermodynamic estimator.

Discretized Hamiltonian $\hat{H}_n$. The polaron Hamiltonian restricted to a uniform n^3 Brillouin-zone grid shared by electron and phonon momenta, closed under momentum addition, with per-arc measure $\frac{1}{N_q}\sum_{\mathbf{q}\neq\Gamma}$, $N_q = n^3$. *This* is what the sampler solves — numerically exactly. All physics statements are statements about the sequence $E_0(n)$.

Thermodynamic limit / $1/n$ extrapolation. Because the discrete BZ sum underweights the $1/q^2$ polar region near Γ, $E_0(n)$ converges from above with leading error $\propto 1/n$. The continuum answer is the intercept of a weighted fit of $E_0(n)$ against $u = 1/n$ (linear plus quadratic term) over the grid ladder $n = 20 \ldots 120$. The spread over fit models is quoted as a systematic. FEP-DMC fits linearly through its three largest grids.

Band mass m_b vs polaron mass m_p. m_b is the Γ curvature of the bare Wannier CBM ($0.6926\, m_e$ for LiF); m_p is the curvature of the polaron dispersion $E_0(\mathbf{p})$. FEP-DMC's mass enhancement "m_p/m_e" means m_p/m_b: its m_e is the band effective mass, not the free-electron mass.

Acoustic-emission (Landau) threshold. The momentum $p_c = m_p c_s$ beyond which the polaron moves faster than sound, so that polaron + one acoustic phonon has lower energy than the quasiparticle at the same total momentum. Beyond it the fixed-τ estimator's $\tau \to \infty$ limit is the continuum bottom, not the quasiparticle energy. For LiF(e), $p_c \approx 0.015$ $\mathrm{Å}^{-1}$, below every nonzero grid momentum.

Central-slice estimator. An observable (phonon number, Z_N, per-mode occupation, electron momentum) measured by inserting its operator at times in a window around $\tau/2$. Unlike the log-derivative energy estimator, whole-worldline averages of occupations are biased by $O(1/\Delta\tau)$ from the ends, where the cloud is still forming.

Remote-band (AHC) correction. The part of the band-edge renormalization outside the sampled Hamiltonian: Fan–Migdal transitions to bands other than the sampled one, plus the Debye–Waller term (absent in any linear-coupling DiagMC). Added perturbatively (Allen–Heine–Cardona); for LiF(e) FEP-DMC finds -0.130 eV.

1.2 Diagrammatics and sampling

DiagMC (diagrammatic Monte Carlo). Stochastic sampling of the Feynman-diagram expansion of an observable — here the imaginary-time worldline expansion of the polaron partition function / Green's function — by a Metropolis Markov chain over diagram topologies, momenta, and times.

Diagram, worldline, backbone. One Monte-Carlo configuration: an electron line (backbone) on $[0, \tau]$, cut into *segments* by interaction vertices, with phonon *arcs* connecting vertex pairs. Stored as a dense τ-sorted array (GPU) or slotmap (CPU).

Segment. A piece of electron line between adjacent vertices, carrying momentum $\mathbf{k}_s$ and weight $e^{-\varepsilon(\mathbf{k}_s)\Delta\tau_s}$.

Arc (phonon arc). A phonon propagator $D(\omega_{\mathbf{q}\nu}, \Delta\tau)$ joining two vertices; carries branch ν and momentum $\mathbf{q}$, which is subtracted from the backbone momentum underneath it.

Vertex (diagrammatic). An electron–phonon interaction event at a time τ_i, weighted by the matrix element $g(\mathbf{k}, \mathbf{q}, \nu)$. Not to be confused with "vertex" as shorthand for the matrix element itself (common in this repo).

Diagram order n, $\langle n \rangle$ (`mean_order`). Number of phonon arcs in the current diagram. The equilibrium mean order tracks coupling strength (LiF at 20^3: $\langle n \rangle \approx 16$); a collapsing or runaway order is the standard symptom of a mis-set μ, a mis-scaled vertex, or under-equilibration.

Update. One Metropolis move proposal (insert/remove arc, shift times, swap, ...). The engine is built from an update set run by a `Runner` through a `MetropolisKernel`.

Chain. One independent Markov chain with its own deterministic RNG stream. Production runs use many chains (12 per CPU node config, 8192 per GPU) for honest, embarrassingly parallel statistics.

Warmup / equilibration transient. Steps discarded before measuring, which must exceed the diagram-order equilibration time ($\sim 10^5$ steps for LiF at 20^3) *per chain* — many short chains do *not* dilute the transient the way one long chain does (the GPU warmup lesson).

Sign, average sign $\langle s \rangle$. The complex vertex makes diagram weights non-positive-definite; each sample carries sign(Re w) and observables are divided by $\langle s \rangle$. For LiF(e) the sign stays benign (≈ 0.99); for the Fröhlich model it is $\equiv 1$.

$|g|^2$-weighted proposal. The native phonon-momentum proposal distribution: new arc momenta are drawn $\propto |g(\mathbf{q})|^2$ (tabulated), so the polar $1/q^2$ peak is proposed where the weight lives instead of rejected.

1.3 Estimators and statistics

Decay estimator. Reads E_0 from the large-τ slope of the measured $G(\tau)$, $E_0 = \mu + \text{slope}$, fitted beyond `tau_fit`. Requires bracketing μ and a fit window clear of excited-state contamination ($\tau_{\text{fit}} \gg 1/\omega_{\text{LO}}$); acceptance criterion is μ-independence of the result. Superseded for production by the thermodynamic estimator; kept as a cross-check.

Thermodynamic (fixed-τ) estimator. The FEP-DMC-style operator estimator: the sampler runs in a fixed-τ ensemble (external time pinned at $\tau_{\max} = \beta$, `fixed_tau: true`) and averages the local energy $E_{\text{loc}} = \tau^{-1}[\sum_{\text{seg}} \varepsilon\,\Delta\tau + \sum_{\text{arcs}} \omega\,\Delta\tau - 2N_{\text{arcs}}]$ over the whole worldline — no fit, no μ bracketing, lower variance. Reported as `e0_thermo`. Systematic error controlled by β (excited-state contamination $\sim e^{-\Delta\beta}$), checked by τ-doubling (guard A).

Exp-reweighted estimator, `energy_estimate`. The older stage-1/stage-2 estimator $\langle -\exp((E_{\text{est}} - \mu)\tau)\rangle$ whose self-consistent update diverges for $\alpha \gtrsim 1$ (fixed-point slope ≈ -2.2); the stage configs freeze `energy_estimate` and it is bisected by hand to the fixed point $E \approx E_{\text{est}}$.

Local energy. The per-worldline value of $\langle \hat{H} \rangle$ decomposed into band, phonon, and interaction accumulators (segment $\varepsilon\Delta\tau$ sums, arc $\omega\Delta\tau$ sums, vertex count); its sign-weighted time average is the thermodynamic estimator.

Thermal phonon propagator. $D(\omega, \tau) = [e^{-\omega\tau} + e^{-\omega(\beta-\tau)}]/(1 - e^{-\omega\beta})$: spontaneous emission + thermal wrap-around, Bose-enhanced. The `zeroTMC` limit hard-codes $\beta \to \infty$, i.e. $D = e^{-\omega\tau}$; LiF(e) production runs are effectively $T = 0$. Modes below a ~ 1 meV frequency cutoff are dropped to avoid the $\omega \to 0$ divergences.

Jackknife. Leave-one-out resampling over measurement batches for error bars, computed as total−batch (never re-aggregating per replicate — the $O(N^2)$ trap). With independent chains the natural batching is one batch per chain.

Batch (`n_batches`). A contiguous block of measurements forming one jackknife unit. Under deadline truncation the batch pooling still guarantees $\geq$ one block per chain.

`steps_done`. The actually executed step count reported by a run; all quoted steps/s throughput numbers use it, so deadline-truncated runs stay honest.

1.4 Ab-initio pipeline (DFT → DFPT → Wannier → vertex)

Quantum ESPRESSO (QE). The plane-wave DFT suite: `pw.x` (SCF/NSCF ground state and band structure), `ph.x` (DFPT phonons). One QE version must be used for the whole `scf/ph/nscf/epw` chain.

DFPT. Density-functional perturbation theory; supplies phonon frequencies and eigenvectors, Born effective charges $\mathbf{Z}^*$, and the electronic dielectric tensor $\boldsymbol{\varepsilon}_\infty$.

EPW. The QE companion code that Wannier-interpolates the electron–phonon coupling; produces the Wannier-representation tensor `LiF.epmatwp` and (with `lpolar=.true.`) subtracts the polar long-range part first.

Wannier functions / interpolation. Localized real-space orbitals obtained from the Bloch states; matrix elements stored on real-space pairs $(\mathbf{R}_e, \mathbf{R}_p)$ interpolate to any $(\mathbf{k}, \mathbf{q})$ by Fourier transform. The electron dispersion $\varepsilon(\mathbf{k})$ used by the sampler is the Wannier-interpolated conduction band (not a parabola).

Disentanglement windows. Wannierization energy windows (`dis_win_max`, `dis_froz_max`) selecting/freezing bands. Must be re-tuned when pseudopotentials shift the band edges, or the frozen CBM slips out.

Pseudopotentials: ultrasoft vs norm-conserving (ONCV). EPW's e–ph matrix elements require *norm-conserving* pseudopotentials; the initial ultrasoft (GBRV) deck silently produced an all-NaN tensor (pitfall P1). The fix (Pseudo Dojo ONCV) raises `ecutwfc` 50 → 96 Ry.

`.epmatwp`. EPW's Wannier-representation e–ph tensor file $g(\mathbf{R}_e, \mathbf{R}_p)$ — the primary input the Rust reader consumes.

`.epb`. EPW's coarse Bloch-representation dump: the *full*, pre-subtraction coupling at the coarse q-points, plus eigenvalues, dynamical matrices, $\mathbf{Z}^*$, $\boldsymbol{\varepsilon}_\infty$. One Fortran unformatted record per MPI pool; the ground truth for the round-trip validation.

`dynq`. The coarse DFPT dynamical matrices; ground truth for the phonon (P7) round-trip.

Vertex $g_{mn\nu}(\mathbf{k}, \mathbf{q})$. The electron–phonon matrix element: scattering of an electron from band n at $\mathbf{k}$ to band m at $\mathbf{k}+\mathbf{q}$ emitting/absorbing a phonon of branch ν, momentum $\mathbf{q}$. Complex; the sampler consumes gauge-invariant $|g|^2$ combinations.

Short/long-range split $g = g^{\mathcal{S}} + g^{\mathcal{L}}$. EPW subtracts the analytic polar (Fröhlich-like, $1/q$) part $g^{\mathcal{L}}$ before Wannierization so the stored tensor $g^{\mathcal{S}}$ is short-ranged; the code re-adds $g^{\mathcal{L}}$ (Verdi–Giustino form, Ewald-summed) exactly, never compressed. The split lives in the Cartesian displacement basis and carries an electron band-rotation factor (pitfall P6).

Polar restoration of the dynamical matrix. The same subtraction applies to the dynamical matrix: the rigid-ion dipole–dipole term (Gonze form, EPW `rgd_blk`) must be added back before diagonalization or the LO–TO splitting is lost and the zero-point factor mis-scales $|g|^2$ by ~ 2 (pitfall P7).

Zero-point factor. The $(\hbar/2M_\kappa\omega_{\mathbf{q}\nu})^{1/2}$ phonon quantization factor in the branch projection of the coupling. Omitting it (pitfall P8) inflated every $|g|$ by $\sim 7\times$.

Acoustic sum rule. The $\mathbf{q} = 0$ self-term subtraction on the diagonal blocks of the long-range dynamical matrix that keeps the acoustic branches gapless at Γ.

Born effective charges $\mathbf{Z}^*_\kappa$, $\boldsymbol{\varepsilon}_\infty$, ε_0. Per-atom dynamical charge tensors and the electronic /static dielectric constants; the ingredients of both long-range terms. Consistency check: Lyddane–Sachs–Teller, $\varepsilon_0/\varepsilon_\infty = (\omega_{\mathrm{LO}}/\omega_{\mathrm{TO}})^2$.

Gauge / band rotation U. Bloch states carry an arbitrary per-$\mathbf{k}$ phase (and mixing, where degenerate); the factor $[U_{\mathbf{k}+\mathbf{q}}U_{\mathbf{k}}^\dagger]_{mn}$ rotates the polar term into the smooth Wannier gauge. Per-band comparisons are only meaningful in gauge-invariant combinations.

***W* star.** The star of the W point of the fcc BZ, contained in the 8^3 coarse grid, where the two lowest conduction bands are exactly degenerate — making per-band round-trip checks gauge-ambiguous there (those q are skipped; the sampler is unaffected).

Reduced model. The single-band, Γ-slice scalar reduction $|g(\Gamma, \mathbf{q}, \nu)|^2$ cached to `reduced-model.json` for the early stage-2 runs; superseded by the full-$\mathbf{k}$ tabulated model but still built by the `model` subcommand.

1.5 Compression and tabulation

SVD compression / rank budget. The short-range Wannier tensor is compressed by per-band-pair truncated SVD; the *global rank budget* r keeps the largest singular values across all band pairs (production: 800 of 1600). A vertex evaluation is then a rank-r contraction $g = \sum_g s_g u_g(\mathbf{k}) v_g(\mathbf{q}, \nu) + g^{\mathcal{L}}$.

Factor tables u, v. The SVD factors tabulated on the momentum grid, stored $\mathbf{k}$-major / $(\mathbf{q}, \nu)$-major so one evaluation streams contiguous memory (the 11–19× cache-layout win); f32 on the GPU tables, with all weight arithmetic f64.

Δ_{rank}. The compression systematic, $E_0(\text{budget}) - E_0(\text{uncompressed})$, measured by two otherwise-identical DMC runs *in the sampler itself* and quoted on every production number. Current certification: $\Delta_{\text{rank}}(800) = 1.0 \pm 0.8$ meV. Never decided by element-wise reconstruction error or the $E^{(2)}$ proxy, and "two nearby budgets agree" is not evidence (convergence in the budget is non-monotonic).

Uncompressed reference. The rank-1600 (full-rank) table used as the comparison baseline for Δ_{rank}; built on demand by `ironmc tables reference`.

Table (`.rmcvtab`). A self-contained ladder artifact (`table-n{grid}-r{rank}-f32.rmcvtab`): the complete discretized Hamiltonian $\hat{H}_n$ — Wannier CBM dispersion, all six polar-restored phonon branches, exact long-range vertex, SVD factor tables, gauge overlaps. DMC runs consume a table; building it is a separate, cheaper step (`ironmc tables build/ladder`).

2 Software engineering

2.1 Validation vocabulary

Validation stages. The general pattern: correctness is certified in ordered stages of increasing physical content (or increasing hardware realism, for the GPU port), each stage mechanical to check, so a failure localizes to the first stage that breaks. The main sequence: engine on analytic Fröhlich → vertex scale → compression convergence → material comparison. The GPU sequence: executable spec → CPU-runtime bit parity → hardware accept-count parity → physics anchor.

Stage. One level of a validation sequence. Concretely, in run configs and the journal: **stage 1** = the engine validated on the analytic Fröhlich model at $\alpha_{\text{LiF}} = 4.76$ against Mishchenko *et al.* (2000) (`configs/stage1-frohlich-lif.json`; sanity anchors $E_0(\alpha{=}1) = -1.013$, $E_0(\alpha{=}2) = -2.08$); **stage 2** = the same engine driven by the *real* LiF tensor, validated against literature ab-initio numbers (`configs/stage2-tensor-lif.json`; Sio–Giustino 2019, and concretely the FEP-DMC 20^3 anchor). "Stage-2-grade number" means: good enough to survive the stage-2 checkpoint, i.e. quotable physics.

Stage-2 checkpoint. The concrete pass/fail criterion of stage 2: $|E_0(20^3) - (-0.2479 \text{ eV})| < 30$ meV against FEP-DMC (`ironmc ladder checkpoint`). Current status: CPU -0.24996 ± 0.00026 eV passes at ~ 2 meV; the first GPU point sat 1.7 meV shallow from under-warmup and is not yet stage-2-grade.

Grid ladder. The D1/D7 production campaign: the sequence of grids $n = 20, 32, 40, 50$ whose $E_0(n)$ feed the $1/n \to 0$ extrapolation. Each grid is one *point* of the ladder ("the 20^3 point"); a repo-wide rename retired the earlier word for this, see *rung* in Section 2.7.

Ladder run. One production DMC run at a single grid point of the ladder: `ironmc ladder run <n> [--rank 800] [--tau120] [--node]` loads the prebuilt `.rmcvtab` table for n^3, runs the fixed-τ thermodynamic-estimator config, and returns a structured record {`grid, rank, tau, e0, stderr, sign, order, steps_done`}. Records pipe into `ironmc ladder fit` for the $1/n$ extrapolation.

Gate. An *encoded*, numeric pass/fail acceptance check that decides a promotion (of a rank budget, a dataset, a backend) — the antidote to eyeballing. The decided gates are cargo aliases / `ironmc gates` commands: `gate-e2` (the $E^{(2)}$ compression gate at 8^3: $|E^{(2)}(800) - \text{ref}| < 3$ meV, $|E^{(2)}(300) - \text{ref}| < 12$ meV, short fraction $< 35\%$) and `gate-dmc` (the in-sampler rank gate, budget vs the uncompressed 1600 reference; scout-sized by default, production via `IRONMC_DMC_STEPS=100000000`). Thresholds are calibrated once, recorded in `journal.md`, then enforced mechanically.

Guard. A gate run *on the production configuration itself* before the expensive sweep: **guard A** (`ladder guard-tau`) doubles $\tau_{\max}$ $60 \to 120$ at 20^3 and requires the E_0 shift $< 2\sigma$ (finite-β systematic under control); **guard B** (`ladder guard-rank`) re-measures $\Delta_{\text{rank}}(800)$ on the production grid with tolerance $\max(2\sigma, 3\text{ meV})$. The remaining points (`ladder sweep`) run only after both guards pass.

Checkpoint. A comparison of a computed number against an external reference with a stated tolerance band (e.g. the stage-2 ± 30 meV band). Unlike a gate it references literature, not an internal baseline.

Smoke test. A deliberately tiny end-to-end run whose only job is to prove the whole path executes and produces sane values before compute is spent — e.g. `ladder-lif-gpu-smoke.json` (256 chains $\times$ 20k steps vs the production 8192×160k + 300k warmup), the first 8^3 DMC smoke, or the reader's physics smoke tests. Institutional memory of pitfall P2: a run can finish cleanly, at the right file size, and be 100% NaN — so smoke tests check *values*, never existence or size.

Pin (pinned value). A hard-coded expected number in a test (band energies, phonon frequencies, tensor dimensions, $\mathbf{Z}^*$ components) asserted against freshly-read data; "reader pins updated" means the dataset changed and the expected values were re-recorded deliberately.

Oracle. An independent, direct (slow, uncompressed) evaluation of a quantity used as the reference in a test — e.g. the direct Wannier-sum vertex against the SVD-tabulated one. Lesson from P8: an oracle that shares a bug with the code under test validates nothing; oracles must be derived independently.

Round-trip. Reconstructing upstream ground truth from the decomposed parts: $g^{\mathcal{S}} + \langle u|u\rangle g^{\mathcal{L}}$ against the coarse `.epb` coupling (machine precision at all 215 coarse $\mathbf{q}$), and short-range + long-range dynamical matrix against `dynq`. The strongest data-preparation check in the repo.

Executable spec. A deliberately flat, fixed-capacity CPU re-implementation of the four sampler updates, simple enough to audit line-by-line, used as the reference the GPU kernel must match bit-for-bit (stages (i)–(ii) of the GPU validation sequence).

Parity. Agreement between two implementations at a stated strictness: *bit parity* (identical reports to the last bit — GPU kernel on the CubeCL CPU runtime vs the flat driver; multi-GPU merge vs single-device run) or *accept-count parity* (identical acceptance counts on real hardware, where lane scheduling forbids bitwise transcript equality). `ironmc gpu parity` runs the hardware check.

Scout-sized vs production. Two sizes of the same measurement: a scout run (minutes, larger error bars) to steer, then the production run (node-hours, quoted statistics) to decide. Gates default to scout size and take an environment override for certification.

Anchor. An external reference number a validation stage lands on (Mishchenko $\alpha = 1, 2$; FEP-DMC -0.2479 eV at 20^3).

2.2 Codebase

`ironmc`**.** This Rust workspace: DiagMC engines and batteries for polaron problems.

`rmc-core`**.** The engine crate: updates, Metropolis kernels, update sets, runner, seeding.

`rmc-diagmc`**.** The state-generic DiagMC engine (runner, updates, decay estimator) behind the `Model` trait.

`Model` **trait.** The abstraction a material implements: electron dispersion $\varepsilon(\mathbf{k})$, phonon dispersion $\omega_{\mathbf{q}\nu}$, complex vertex $g(\mathbf{k}, \mathbf{q}, \nu)$, and a phonon-momentum proposal. Two implementations: analytic Fröhlich and the ab-initio LiF model.

`rmc-frohlich`**.** The standalone Fröhlich polaron engine and performance fixture.

`rmc-ab-initio`**.** The ab-initio consumer app: EPW readers, Wannier reconstruction, polar terms, SVD tabulation, table building, and the production run/merge CLI.

`rmc-minimal`**.** The hot-path benchmark fixture crate.

Config (`configs/*.json`**).** A run definition: sampler knobs (`mu`, `max_tau`, `chains`, `max_steps`, `warmup_steps`, `n_batches`, `seed`, `fixed_tau`, ...). Naming: `stage1-*/stage2-*` for the validation stages; `ladder-lif*` for grid-ladder production (`-node` = full-node chain count, `-tau120` = guard-A variant, `-gpu` = GPU population, `-smoke` = smoke-sized).

Report. The JSON a run emits (`e0_thermo`, `sign`, `mean_order`, `steps_done`, ...), consumed as structured data by the `nu` tooling.

`RankDump / merge`**.** Per-SLURM-rank raw measurement dump written by multi-GPU jobs (default under `results-gpu/<jobid>`), merged offline into one report; the merge is bit-identical to the equivalent single-device run by the global-chain-id construction.

2.3 Reproducibility and RNG

Deterministic chain streams. The load-bearing invariant: each chain's RNG stream is derived from (master seed, *stable* chain id), never from scheduling order — so thread/rayon scheduling can never change which stream a chain gets. CPU: xoshiro256++ streams. GPU: counter-based Philox4x32-10 keyed by (seed, chain, step), which lets all 32 lanes of a warp compute identical draws with no communication.

Global chain id. Chain ids are global across ranks (offset = rank $\times$ chains-per-rank), so a multi-rank population is a pure repacking of a single larger run and merged statistics are a function of (seed, total chains) only.

f64 policy. All physics arithmetic is f64 (weights, τ, energies); f32 appears only as a *storage* format for the big factor tables. Momenta are exact integer grid indices.

2.4 GPU port

CubeCL. The Rust GPU compute framework used for the CUDA backend; its CPU runtime executes the same kernel on the host, enabling the bit-parity stage.

Warp-per-chain / redundant lanes. The kernel's parallel mapping: one chain per 32-lane warp, every lane redundantly executing the chain's update logic (warp-uniform control flow for free), lanes cooperating only inside vertex evaluations.

Lockstep updates. All lanes of a chain's warp (and, under group-uniform selection, all chains of a group) attempt the same update type each step. Mandatory on device: barrier arrivals pair by dynamic count, so divergent lanes either deadlock or silently corrupt the shared scratch memory — and the barrier count per step is state-dependent, so uniformity must hold over the whole dynamic trace. Statistically free: the update-type schedule is state-independent, so each chain's marginal law is the standard random-scan chain (selection probabilities cancel in the acceptance ratio as usual); a shared schedule only correlates chains weakly — variance, never bias. Corollary: impossible proposals count as rejections, never redrawn (a state-dependent redraw would bias the run).

Cooperative vertex dot. The rank-r contraction strided across the 32 lanes with a shared-memory reduce; the emit/absorb pair $g(\mathbf{k}_e, \mathbf{q}, \nu)\, g^*(\mathbf{k}_a, \mathbf{q}, \nu)$ fuses both dots over one shared $v_g(\mathbf{q}, \nu)$ strip, halving table traffic.

Bandwidth-bound. The regime the design targets: a vertex evaluation streams $\approx 16r$ bytes, so HBM bandwidth (≈ 1.24 TB/s sustained on an A100, $\approx 80\%$ of peak) — not FLOPs — sets throughput; this is where a GPU beats a CPU node by an order of magnitude, and why the margin grows with grid size once the table no longer fits CPU L3.

Planar layout. Complex tables stored as separate real/imaginary f32 planes for coalesced loads.

`ironmc gpu` **commands.** `run` (production GPU run), `scale` (chain-count throughput sweep), `parity` (hardware parity test), `vbench` (vertex-evaluation microbenchmark), `info`.

2.5 Cluster operations

VSC-5. The Vienna Scientific Cluster the campaigns run on: 128-core Zen3 CPU nodes (512 GB–2 TB) and 2×A100 GPU nodes, driven via SLURM.

Partition / QoS / devel. SLURM node classes (`zen3_0512`, `zen3_0512_a100x2`, ...) with default and *devel* QoS — the devel queue gives fast turnaround at a fixed 10-minute walltime cap; smoke- and scout-sized runs are shaped to fit it. The partition table (including GPU counts) lives in `nu/ironmc/slurm.nu`.

`ironmc slurm submit`**.** Wraps a target command (e.g. `ladder run 50`) in a generated per-job SLURM script with its own job directory and results under `./results/`.

Deadline truncation. Sizing `max_steps` beyond the walltime and letting the SLURM end-time stop the run; statistics remain valid because batches pool across chains and throughput is quoted from `steps_done`.

`--node` **config.** The full-node CPU variant of a ladder run: chain count auto-sized to the CPU allocation, 10^7 steps per chain.

2.6 Tooling and process

`rtk`. Token-optimizing CLI proxy; shell commands in agent sessions are prefixed with `rtk` (e.g. `rtk git status`).

`nu/ironmc`. The Nushell operations module (`ladder`, `tables`, `gates`, `gpu`, `slurm`, `qe`, `bench`, `test` submodules); every command returns structured records meant to be piped, not parsed.

`journal.md`. The append-only lab notebook: dated entries recording each implementation chunk, calibration, gate result, and lesson. Methodology graduates from the journal into the LaTeX note.

Plans (`plans/{open,active,done}/*.md`). Design documents for work campaigns (`plan-large-grids.md`, `gpu-abinitio.md`, ...), sorted into status subdirectories (open / partly done / implemented). Steps inside a plan are labeled `A0, A1, ...` (phases), so "A0" or "D7" always refers to a step *of a specific plan*; the journal names the plan when ambiguous.

G/N/D/P labels. The methodology note's own numbering: **G**oals, **N**on-goals, **D**eferred items, **P**itfalls (P1–P9). Retired non-goals keep their labels for reference.

Phase A / Phase B. The two halves of the original ab-initio plan: A = data preparation (QE/EPW runs, readers, reconstruction), B = the DiagMC consumer (sampling the prepared vertex).

Profiles. Cargo build profiles: `release` is the default for everything (debug is off limits without a stated reason); `release-tuned` exists for tuned performance builds.

`bench-compare`. The benchmark harness alias comparing steps/s across revisions with repetitions; benchmarks need an isolated core and are run by the user, not by agents.

2.7 Overloaded terms — read by context

rank. (1) SVD rank budget of the compressed vertex (800, 1600) — the sense in `--rank`, `gate-dmc`, Δ_{rank}, table filenames. (2) SLURM/MPI rank — one process, one GPU, in multi-device jobs (`RankDump`, "two-rank job"). Table names use sense (1); results directories use sense (2).

rung (retired). Historical term, removed from the working tree on 2026-09-11 but surviving in git history and commit messages. It entered via the `plan-rmc-ab-initio.md` "validation ladder" metaphor and ended up carrying two senses at once — a validation *stage* ("rung 1" = Fröhlich engine check, "rung 2" = real-tensor check, "rung-2 checkpoint") and a grid-ladder *point* ("rung-1 run" in throughput contexts = the 20^3 production run) — which is why it was retired. Reading old history: validation contexts → *stage*, grid contexts → *point*.

ladder. Means the grid ladder only (the $1/n$ extrapolation campaign; the `ironmc ladder` command family). Pre-rename documents also said "validation ladder" for what is now *validation stages*. For external writing, prefer "grid-convergence sequence": in many-body theory "ladder diagrams" and the ladder approximation are established, unrelated terms.

vertex. (1) The e–ph matrix element $g(\mathbf{k}, \mathbf{q}, \nu)$ ("vertex evaluation", "vertex cache"). (2) A diagrammatic interaction event on the worldline ("vertex count N_v").

order. (1) Diagram order (number of phonon arcs). (2) Perturbative order ($E^{(2)}$ is second order in g). (3) Ordering of τ values in the diagram array ("τ-sorted").

table. (1) The `.rmcvtab` discretized-Hamiltonian artifact. (2) The in-memory SVD factor tables. (3) Nushell's tabular output.